

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm

(Clinical Pharmacy & Pharmacology and Toxicology)

University Theory Examination January 2018

PHCC004 ANALYTICAL TECHNIQUES

Date: 01.01.2018 , Monday

Time: 10.00 a.m. to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Enlist the UV spectrophotometric methods for multicomponent analysis. Write a note on any one of them. 05
- B) Enlist various ionizers used in Mass Spectroscopy. Write a note on MALDI. 05
- Q.2 A) What is NMR? Write a note on factors affecting chemical shift. 05
- B) Write a note on sampling techniques used in IR spectroscopy. 05
- OR**
- Q.2 A) Discuss the advantages of UV spectrophotometric method over separation techniques. Write a note on solvents used in UV spectrophotometric methods. 05
- B) Enlist factors affecting fluorescence spectroscopy. Discuss the instrumentation and applications of spectrofluorimeter. 05
- Q.3 Answer the following (Any three) 15
- (A) Discuss in brief cell imaging techniques.
- (B) Write a note on flow cytometry.
- (C) Write a note on fragmentation rules in mass spectroscopy.
- (D) Write a note on FT-NMR technique.

SECTION-II

- Q.4 (A) What is chromatography? Discuss stationary phases used in GC. 05
- (B) Discuss the principle of Ion exchange and Size exclusion chromatography. 05
- Q.5 (A) Explain: LC-MS, Flash chromatography 05
- (B) Differentiate HPLC and HPTLC. Write applications of HPTLC. 05
- OR**
- Q.5 What is Isoelectric focusing? Discuss in brief. Write applications of electrophoresis. 10
- Q.6 Answer the following (Any three) 15
- (A) Write note on RIA.
- (B) Write brief note on Immunofluorescent microscopy.
- (C) Discuss in situ gene expression technique.
- (D) Discuss capillary electrophoresis.
